

EXHIBIT 4

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLORADO
Honorable Marcia S. Krieger**

Civil Action No. 09-cv-00019-MSK-MEH

UNITED STATES OF AMERICA,

Plaintiff,

v.

CEMEX, INC.,

Defendant.

**DEFENDANT CEMEX, INC. RESPONSES TO UNITED STATES' FIRST SET OF
INTERROGATORIES**

Pursuant to Rule 33 of the Federal Rules of Civil Procedure, Defendant CEMEX, INC.
("CEMEX") responds to Plaintiff United States of America's ("Plaintiff's") First Set of
Interrogatories as follows:

PRELIMINARY STATEMENT

CEMEX is in the process of diligently conducting its investigation of the facts pertaining to this action, while responding to the Plaintiff's discovery requests. That response has entailed a time-consuming document search (that has already produced more than 80 boxes of documents and thousands of pages of electronic materials) as well as the research needed to respond to these interrogatories. CEMEX has not yet been able to obtain any information from formal discovery in this action. Accordingly, the responses and objections herein are based on the information presently known to CEMEX, after a diligent search of the information within its possession,

and Identify, General Instruction (F) supra, the Documents in which this information is found and each person with knowledge pertaining to the information requested by this Interrogatory:

- a. the date upon which construction of the change began, the date upon which construction of the change was completed, and the first date upon which the modified unit commenced operation;
- b. the nature and extent of the change, including a description of how the design, physical characteristics, and/or operational characteristics of the modified unit changed;
- c. the cost of the change;
- d. all pre- and post-change reports, critiques, audits or other documents that discuss how the project was performed;
- e. unit throughput pre- and post- change;
- f. number of hours of operation pre- and post-change;
- g. actual emissions of NOx and PM-10 in tons per year, tons per month and pounds per hour emitted pre- and post-change;
- h. potential to emit of NOx and PM-10 in tons per year, tons per month and pounds per hour emitted pre- and post-change from the modified unit; and
- i. Impact on clinker production capacity.

RESPONSE TO INTERROGATORY NO. 5:

CEMEX objects to this interrogatory on the grounds that it is overly broad, unduly burdensome and seeks information neither relevant to this litigation nor reasonably calculated to lead to the discovery of admissible evidence. In particular, to “Describe In Full and Complete Detail, each and every structural or operational change(s)” to the identified units would be a

massive undertaking, and the inclusiveness of the terms “structural and operational changes” is both uncertain and could include minutiae. CEMEX also objects to this request as vague and ambiguous as to the terms “ancillary systems,” and “operational characteristics.” CEMEX also objects to Interrogatory No. 5 to the extent that it seeks production of attorney-client privileged information, attorney work product, or non-discoverable expert information. The analysis of relevant “structural and operation change(s)” and their impacts will include all of these categories of information. The methodology and calculations on which this contention will be based will be developed by experts, from whom the United States may obtain discovery exclusively under Rule 26(a)(2) and Rule 26(b)(4) of the Federal Rules of Civil Procedure. The Documents which contain factual information responsive to Interrogatory No. 5 have already been produced or are in the process of being produced to the United States in CEMEX’s response to the United States’ first request for production of documents, and in CEMEX’s response to U.S. EPA’s demands for information under Section 114 of the Clean Air Act. Finally, CEMEX objects to this interrogatory to the extent that it is duplicative or cumulative of other requests.

Subject to and without waiving its foregoing General Objections and specific objections, and in accordance with Mr. John Moscato’s letter to Chet Thompson dated August 17, 2009, CEMEX responds as follows:

With respect to projects undertaken at the clinker cooler, finish mill circuit, raw feed preparation circuit, and oxygen plant between 1998 and 2000, CEMEX incorporates by reference and refers the United States to CEMEX’s response to Request #3 of EPA’s June 6, 2007 Clean Air Act Section 114 Information Request and attached information.

Listed below are structural or operational changes made to the Raw Mill System, Kiln System, Finish Mill System, and/or Oxygen Plant between 1995 and 2009 with the potential to impact clinker production positively (*i.e.*, increase production) or negatively (*i.e.*, decrease production). CEMEX is still investigating the details of these projects. Moreover, the potential impacts of these projects on both production and emissions to the atmosphere will likely be the subject of expert testimony, and hence the appropriate process for the United States to seek additional discovery is through Federal Rule of Civil Procedure 26(a)(2) and (b)(4).

Raw Mill System

- Added a 3rd air compressor for material conveying and to stabilize production during cold and damp weather (circa 1997)
- Material transport line was increased from 8” to 10” to compensate for additional back-pressure from 3rd compressor (circa 1998)
- Mill discharge screen changed from 7/16” to 5/8” slots to address reduced mill efficiency due to increased re-circulating load (circa 2002)
- Power band on pump drive was added to reduce pump drive slipping (circa 1996)
- Added 3 dust collectors on top of raw silos to vent silos (circa 1995)
- Two compartments of baghouse 325-1 taken out of service due to addition of dust collectors (circa 1998)
- Varying amounts of purchased limestone were used in the finished mill to meet ASTM standards, fed via the outside hopper instead of from the raw silo diversion gate (circa 2005)

Kiln System

- Various changes in feed chemistry, including:

- o Multiple sources of iron, including iron ore, slag, magnetite, mill scale (circa 1998 to 2007)
 - o Dowe Flats silica trial was performed but discontinued due to high alkali content (circa 2002)
 - o Fly-ash as an alumina source added to the raw mill (implemented circa 2001 and discontinued in 2004)
 - o Calcium Chloride was added to feed table to assist in production of low alkali clinker (circa 1998 and 2008)
 - o Change in limestone source from Larimer to Dowe Flats (circa 1999)
- Changed the percentage of calcined material returned to the kiln:
 - o Returned alkali bypass dust generated from the by-pass fan to the kiln to reduce landfilling of calcined dust (circa 2001)
 - o Shut down by-pass fan to decrease dust loss and power consumption circa 2004, but ran trials 1998-2002; periodically operated system since 2004
 - o Extended thimble in cyclone to reduce calcined dust loss (circa 2006)
- Improved efficiency of fuel firing and adjustments to deal with changes in coal quality
- The refractory thickness in certain zones of the preheater tower combustion chamber was increased to increase air velocity through the combustion chamber, which improved the mixing of the fuel and oxygen, resulting in reduced overburn (combustion that is delayed to beyond the combustion zone)
- Pyro-processing modifications caused by coal quality resulted in operational adjustments and production impacts. Coal quality parameters include but are not limited to the following:
 1. Moisture content, which effects grindability and flow

2. Sulfur content, which can effect build-up in the kiln system
 3. Ash percentage and mineralogy, which effects raw mix chemistry and burnability
 4. BTU content, which effects burnability
 5. Volatility and fineness, which aides ignition and promotes combustion
 6. Hargrove index (hardness of coal), which effects grindability
- Coal pipe booster fan and corresponding burner pipe were modified to change flame shape and efficiency (circa 2000)
 - Repaired the coal feed line to the preheater tower with differing size pipes to improve complete combustion of coal in 2000 (to 24”), in 2004 (to 6”), and in 2006 (to 10”)
 - Maxon heater was added to aid drying of coal to coal mills in winter and wet months (circa 1995)
 - Larger Maxon heater was added to enable kiln coal use during pre-heating periods (circa 2007)
 - Added a transport line, “Y” and valve to kiln feed silo to enable kiln feed to be transported to either the kiln feed silo or the homogenizing silo, rather just the homogenizing silo to improve blending (circa 1996)
 - Repaired kiln motor increasing the horsepower but not changing the fan curve (circa 1996)
 - The feed pipe to the kiln silo was increased in diameter from 8” to 10” in order to reduce the backpressure on the pumps (circa 1998)
 - Tertiary duct changes included:
 1. Modify drop out discharge chute to reduce plugging (circa 1999)
 2. Modify drop out discharge chute to further reduce plugging (circa 2008)

3. Air blasters in swirl chamber and inlet duct were added to reduce build-up (circa 2008)
 4. Air blasters on take-off duct were added to reduce build up (circa 2008)
- Whole tire firing test was performed to test the use of alternative fuels (circa 2003)
 - Larger motor on clinker breaker was installed to prevent plug up (circa 2008)
 - New dust collector over outside clinker pit vents to IKN cooler fan inlet was installed (circa 2004)

Finish Mill System

- High efficiency separator installed and corresponding mill ball charge was adjusted to improve mill efficiency and meet higher cement fineness demands (circa 1998)
- Rejects from separator discharge were relocated to the elevator above the boot versus into the boot to prevent possible back-spilling (circa 2005)
- Larger motor on feed elevator drive was installed to increase re-circulating load and mill efficiency (circa 2008)

INTERROGATORY NO. 6:

Beginning on January 1, 1995 and continuing to the present, Identify, General Instruction F supra, and Describe In Full and Complete Detail, the hourly, daily and monthly mass of emissions of NO_x and PM-10 from the Raw Mill System, the Kiln System, the Finish Mill System and the Oxygen Plant (including any emissions from associated and/or ancillary systems and/or equipment like, e.g. a calciner or bag house). In Your Response, Identify and Describe In Full and Complete Detail how this information was generated, gathered, and stored (including the format it is stored in), Identify the Documents where this information can be